

Dhruv Arora

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EDUCATION

University of Wisconsin-Madison, BS. Computer Science & Math

Expected May 2028

- Activities: Traders @ Wisconsin, Alpha Kappa Psi Business Fraternity, Formula SAE, UW Climbing Team, ADZ Event Manager

PROFESSIONAL EXPERIENCE

Autodesk, Software Engineering Intern & Head Marketing Intern

Portland, OR

React, Typescript, Docker, Three.js, Python, Jira, PowerPoint, AWS, JSON, ESLint, Biome, Fusion June 2024 – August 2026

- Developed & marketed Synthesis, Autodesk's open-source robotics simulator, across 3-year tenure; grew users from 5K to 20K+
- Led user research Zoom sessions with robotics teams to drive product iteration; led team pitch to Autodesk EVP & executives

SwiftLabs, Partner/Developer

Clive, IA

React, Express, Biome, Vite, Vercel, Linear, Git, Github, Firebase, Tailwind CSS

June 2025 – Present

- Built 5-figure government contract for *School Administrators of Iowa*; assist principals with drafting teacher evaluation reports
- Designed and deployed fully-functioning evaluation portal to 2K+ government-authorized users, interfacing directly with clients

True Martial Arts, Senior Instructor

Sammamish, WA

Instructing, Word, Excel, Performance Evaluation, Team Coordination, Planning

April 2021 – June 2025

- Trained 30+ Assistant and Head Instructors & oversaw promotions; directly evaluated 100+ students in Taekwondo proficiency
- Organized student events and tournaments for 700+ participants from 4 martial arts schools, managing judges, scoring, & safety

PROJECTS

Orion, Metal, C++20, Slang, SwiftUI, LibRaw, CMake, Ninja, Little-CMS

June 2026

- Built open-sourced GPU-accelerated raw photo editor for macOS; 27-node compute DAG re-renders RAWs in 8ms full-res
- Ported 8+ research published algorithms incl. RCD demosaic, wavelet denoise & AgX transform; 4,600+ GPU validated unit test

Nazar, Python, FastAPI, Ollama, networkx, React, Bash, Vite, Tailwind CSS, SSE

March 2026

- Won 2nd of 40 teams at Dell x NVIDIA hackathon; built air-gapped AI agent diagnosing offline server faults via local 122B LLM
- Engineered agentic loop pairing BM25 search & networkx graph traversal to trace faults keyword search alone cannot connect

Asclepion, React, Typescript, Pytorch, Python, Ollama, Sigma.js, Uvicorn

February 2026

- Built clinical reasoning assistant using local LLM-powered graph visualization to map patient symptoms to potential diagnoses
- Trained PyTorch BERT model on medical billing data to score diagnosis-to-item relevance; generates pdf bill from patient notes

Wisconsin Racing, C, Datasheets, GPIO, ESP32, STM32 IDE, Reflow soldering, Slack

September 2025 – Present

- Wrote embedded system that implemented cell balancing, over/under-voltage protection for one-third-scale Formula 1 track car
- Built GPIO driver for ESP32 chip to communicate with ECU of the car and translate data into a format displayable on a 5" screen

Soundwave, Tensorflow, Numpy, Pandas, MATLAB

November 2023 – February 2025

- Secured \$150K Letter of Intent from Konexions building proprietary TensorFlow model using MFCCs to localize foreign accents
- Placed 1st globally for *TiE Young Entrepreneurs* comp earning \$7K and 1st nationally for the *Diamond Challenge* earning \$12K

Princeton AI4ALL, Pytorch, Python, Jupyter Notebook, CUDA, Slack

July 2023 – August 2023

- Built and trained a PyTorch model for satellite-image deforestation classification at Princeton, using Google colab TPU workflow
- Presented findings in Washington, DC to the FTC and CFPB, and the Cooley law firm, discussing policy implications of research

LEADERSHIP

Alpha Kappa Psi, Administration Committee & Member

September 2025 – Present

- Automated attendance for 150 brothers via self-hosted Linux server syncing event & form CSV data into master Google Sheet
- Represented chapter at recruiting events incl. New York outreach trip; onboarded 400+ rushes onto automated attendance sheet

DECA, Chapter Vice President

May 2024 – June 2025

- Managed communications for 600+ members; coordinated travel logistics enabling 40+ to attend the international competition
- Led DECA Inc. campaign initiatives chapter-wide; directly mentored 20+ members in cold outreach efforts to business leaders

AWARDS

Hackathons: 2nd Place, Dell x NVIDIA Hackathon (BuilderBase, 2026); 3rd Place, CascadiaJS Hackathon (Apify, Box & AWS, 2025)
Competitions & Honors: 3rd Place, Congressional App Challenge (2024); Top 10 International Finalist, DECA (2023); Worlds Qualifier, FIRST Robotics (2021, 2025); Eagle Scout, Order of the Arrow (2023); 2nd Degree Black Belt, Taekwondo & Arnis (2025)

SKILLS

Interests: classical piano, trail building, microphone architecture & acoustic engineering, bouldering, ice climbing, mountaineering

Languages: C, C++, Rust, Python, Java, C#, Dart, R, MATLAB, TypeScript, JavaScript, HTML, CSS, Bash, LaTeX, JSON, TOML |

Frameworks: React, Next.js, Node.js, Express, Flask, Flutter, JavaFX, WebGL, Tailwind CSS, PyTorch, TensorFlow, NumPy, Pandas, SciPy |

Tools: Git, GitHub, Linux, AWS, MySQL, Arduino, npm, Cargo, Gradle, Maven, Vite, ESLint, Prettier, Biome, Jira, MS Office